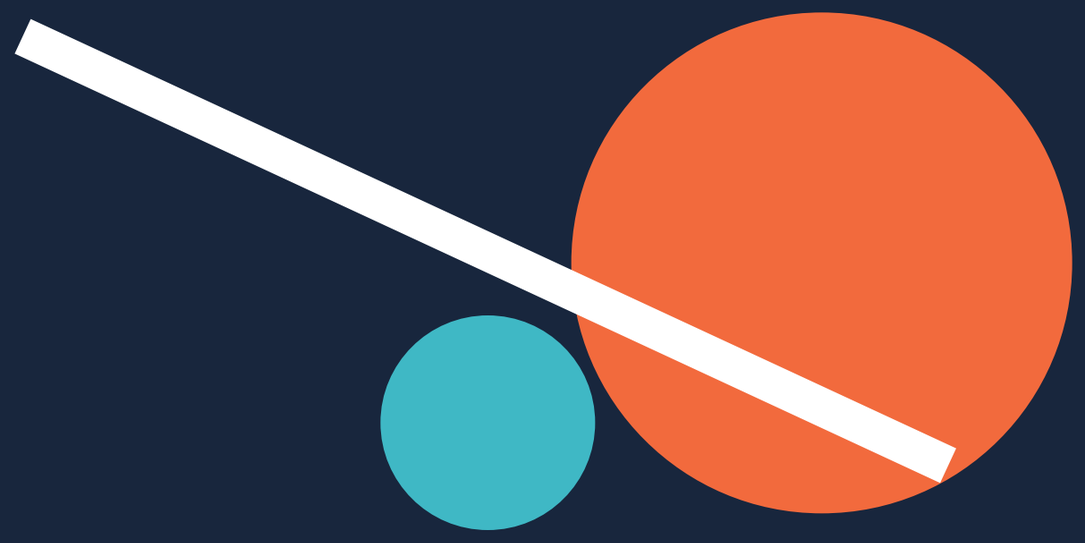
A RESOURCE-LIGHT RAG BENCHMARK

Does document structure improve RAG?

AutoSTAIR compares ordinary chunk retrieval with an index-guided pipeline on one long public report.



THE QUESTION

A document already has a map. Should RAG use it?

Most RAG pipelines split text into equal-sized chunks and search everything at once. Long reports and books already contain human-authored structure: chapters, headings, and an index.

Objective

Measure whether routing through that structure improves retrieval and final answer quality under a controlled setup.

RESEARCH INSPIRATION

IBM Research's STAIR paper

STAIR predicts the relevant leaf section from a question and the document's full table of contents. It also introduces SearchTome, a benchmark built from 18 books across six domains.

Our project asks a smaller, practical question. We use existing document structure directly, without training a retriever or generating a new hierarchy.

Inspired by the paper, not a reproduction of it.

THE TEST BED

One public report. Two retrieval routes.

276
PAGES

395
CHUNKS

54
AUTHORED SECTIONS

20
QUESTIONS

Source: World Bank, World Development Report 2024: The Middle-Income Trap.

CONTROLLED DESIGN

Same evidence and models. Different route.

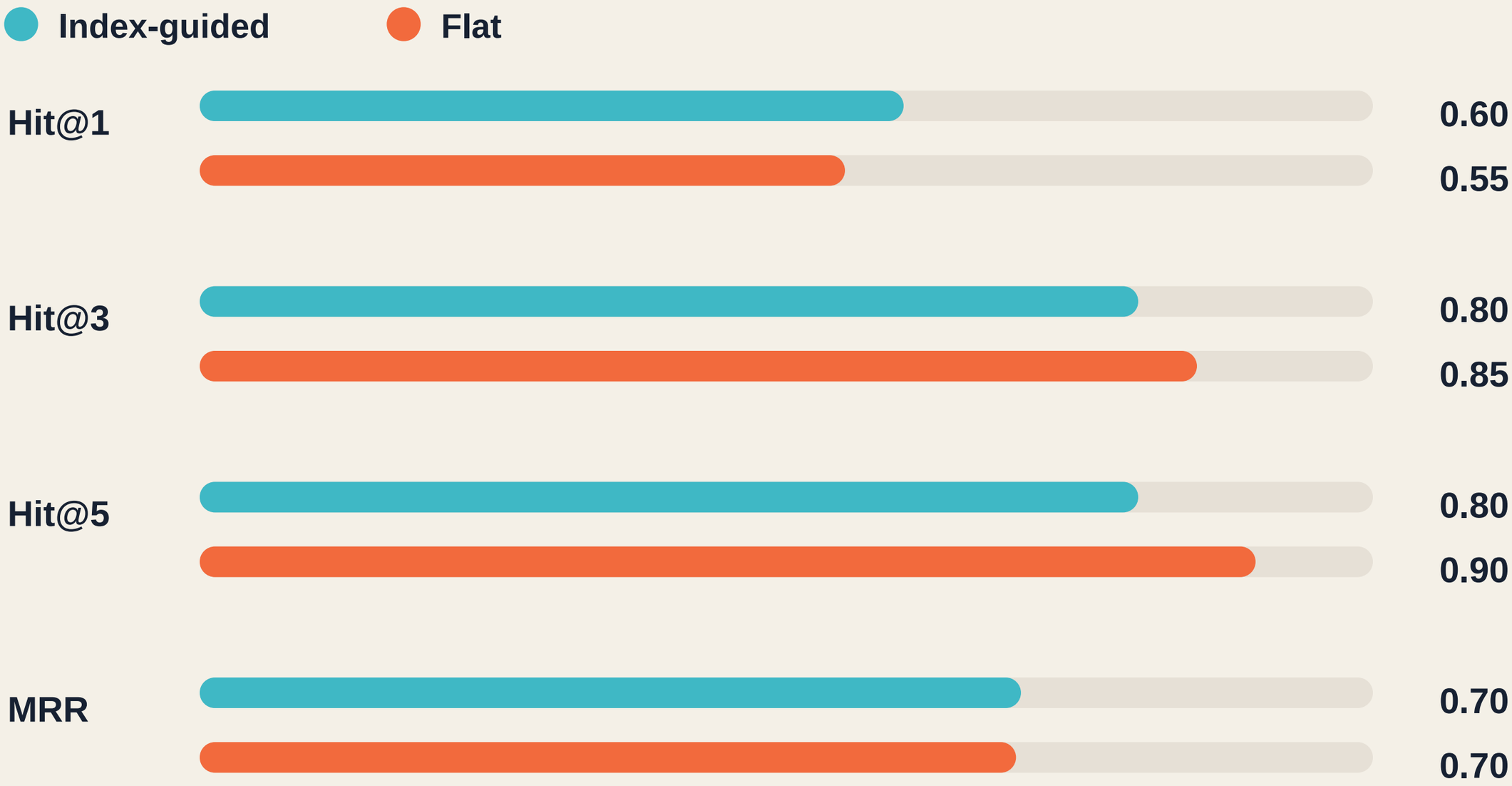
FLAT RAG

Question -> search all 395 passages -> answer

INDEX-GUIDED RAG

Question -> select 1 of 54 sections -> search within it -> answer

Structure helped at rank one.



Flat search recovered more relevant evidence at depth: Hit@5 reached 0.90 vs 0.80.

MANUAL ANSWER REVIEW

More wins did not mean a higher average score.

8	6	6
INDEX-GUIDED WINS	FLAT WINS	TIES

4.85 / 5

FLAT OVERALL QUALITY

4.61 / 5

INDEX-GUIDED QUALITY

Blinded review: correctness, grounding, completeness, and citation quality.

WHY THE RESULT IS MIXED

The section decision can become a bottleneck.

WHEN ROUTING WORKED

The evidence was focused, direct, and easier to audit.

WHEN ROUTING FAILED

The correct passage was outside the selected section, so the second stage could not recover it.

Two cross-section questions produced safe abstentions instead of answers.

A benchmark of layouts, not a trained retriever.

No fine-tuning

We compare practical retrieval pipelines using existing embeddings and generation.

One public report

STAIR evaluates 18 books across six domains; this is a deliberately small case study.

End-to-end RAG

We examine both retrieval metrics and the quality of cited answers.

Existing structure

We preserve the report's authored sections instead of creating a hierarchy with AI.

MAIN TAKEAWAY

Use structure as a strong routing signal, with a global fallback.

A hybrid design can keep the focus of index-guided retrieval without losing evidence when the first section choice is wrong.

Small study. Useful signal. Clear next experiment.

FULL METHODOLOGY, CODE, AND RESULTS IN THE POST